Aleksas Mazeliauskas

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Professional experience

2019–2021 Senior research fellow, **Theoretical Physics Department, CERN**

Research on heavy-ion phenomenology, QCD thermalization and future ion experiments at the LHC.

2017–2019 Postdoctoral researcher, **Institute for Theoretical Physics, Heidelberg University**

Member of collaborative research center SFB 1225 *Isolated quantum systems and universality in extreme conditions*.

Work on non-equilibrium QCD physics using quark and gluon kinetic theory, pre-equilibrium kinetic propagator `KøMPøST`, resonance decay code `FastReso`.

2014–2017 Research assistant, **Nuclear Theory Group, Stony Brook University**

PhD dissertation work under supervision of Prof. Derek Teaney, *Fluctuations in ultra-relativistic heavy ion collisions*

Education

2012–2017 **Stony Brook University**, United States, Doctor of Philosophy in Physics
Max Dresden Prize for outstanding theoretical thesis, degree awarded on 19/05/2017

2011–2012 **University of Cambridge**, United Kingdom, Master of Mathematics (Part III Maths)
graduated with distinction

2008–2011 **University of Cambridge**, United Kingdom, Bachelor of Arts (Mathematical Tripos)
1st class degree

Awards

09/11/2019 Nuclear Physics A Young Scientist Award for the best theory talk at Quark Matter 2019

02/05/2017 Max Dresden Prize for outstanding theoretical thesis

25/07/2016 APS FGSA Travel Award for Excellence in Graduate Research

23/05/2013 David Fox award for the outstanding Teaching Assistant

Professional services

Events organised

2019 Convener of discussion group on "Connections between QGP-like observables in small systems" at **International Workshop on QCD Challenges from pp to AA**, 19-23 Aug 2019, Lund, Sweden, and the editor of the summary report.

Convener of discussion group on "Implementations of critical dynamics" at **EMMI Rapid Reaction Task Force: Dynamics of critical fluctuations**, 8-12 April 2019, Darmstadt, Germany, and the editor of the summary report.

Refereeing

Physical Review Letters, Physical Review C and D, Physics Letters B, Nuclear Physics B

Languages

Lithuanian	native, enthusiastic teacher	**French**	elementary (A2)
English	fluent, higher education in English	**Russian**	basic conversational
German	upper intermediate (B2)	**Chinese**	beginner, passed HSK level II test

Programming

Programming	Extensive numerically oriented coding in `C++` and `Python`, previous experience with `Fortran` and Matlab, good working knowledge of Mathematica
General IT	Git (source code management), LaTeX (word processing), Inkscape (vector graphics), GNU/Linux (operating system), Gnuplot (graphing), HTML5 (web building)

Invited presentations

24/09/2020	talk at **ALICE Physics Week (online)** on *Partonic rescattering in light nucleus collisions*, Bucharest, Hungary
28/05/2020	talk at **VIII Large Hadron Collider Physics Conference (online)** on *Equilibration of QGP in small systems*, Paris, France
06/12/2019	talk at **XIX Zimányi School Winter Workshop** on *Pre-equilibrium phenomena in Quark Gluon Plasma*, Budapest, Hungary
25/11/2019	talk at **Theoretical Foundations of Relativistic Hydrodynamics Workshop** on *Pre-scaling, hydrodynamic attractors and entropy production in heavy ion collisions*, Banff, Canada
01/11/2019	talk at **New Development of Hydrodynamics and its applications in Heavy-ion Collisions Workshop** on *Hydrodynamic attractors, initial state energy and particle production in relativistic nuclear collisions*, Shanghai, China
13/09/2019	talk at **XLIX International Symposium on Multiparticle Dynamics** on *Equilibration in Quark Gluon Plasma*, Santa Fe, New Mexico, USA
25/06/2019	plenary talk at **V Initial Stages Conference** on *Matching of initial conditions to hydro evolution*, New York, USA
06/04/2019	talk at **XIV Polish Workshop on Relativistic Heavy-Ion Collisions** on *Chemical equilibration in hadronic collisions*, Krakow, Poland
01/08/2018	parallel talk at **XIII Quark Confinement and the Hadron Spectrum Conference**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, Dublin, Ireland
18/05/2018	plenary talk at **XXVII Quark Matter Conference** on *Initial conditions for nuclear collisions: theory overview*, Venice, Italy
02/10/2017	talk at **XII High-pT Physics Workshop** on *Initial conditions for heavy ion collisions* , Bergen, Norway
15/06/2017	talk at **XII Workshop on Particle Correlations and Femtoscopy** on *Kinetic theory equilibration for realistic heavy ion initial conditions*, Amsterdam, The Netherlands
06/03/2015	talk at **Collectivity in Small Colliding Systems with High Multiplicity Workshop** on *Principal Component Analysis and Subleading Flow*, Brookhaven National Lab, New York, United States

Contributed presentations

03/06/2020	**X Hard Probes (online)**, Austin, USA
06/11/2019	**XXVIII Quark Matter Conference**, Wuhan, China, BEST THEORY TALK
22/09/2019	**Quantum Systems in Extreme Conditions**, Heidelberg, Germany
10/06/2019	**XVIII Strangness in Quark Matter**, Bari, Italy
14/01/2019	**XLVII Workshop on Gross Properties of Nuclei and Nuclear Excitations**, Hirschegg, Austria
23/05/2019	**Origins of Correlations in High Energy Collisions**, INT, Seattle, Washington, United States
09/04/2019	**EMMI RRTF: Dynamics of critical fluctuations**, Darmstadt, Germany
02/10/2018	**IX Hard Probes conference**, Aix-Les-Bains, France
27/08/2018	**Probing the Quark-Gluon Plasma with collective phenomena and heavy quarks**, MIAPP, Munich, Germany
19/09/2017	**IV Initial Stages**, Krakow, Poland
08/08/2017	**Critical Point and Onset of Deconfinement**, Stony Brook, New York, United States
07/02/2017	**XXVI Quark Matter**, Chicago, Illinois, United States
19/09/2016	**VII Hot Quarks**, South Padre Island, Texas, United States
25/05/2016	**III Initial Stages**, Lisbon, Portugal
22/07/2015	**Correlations and Fluctuations in p+A and A+A Collisions**, INT, Seattle, Washington, United States
27/03/2015	**Ohio-Region APS meeting**, Kent, Ohio, United States

Publications

My publication record consists of 19 articles, 1 review, 2 community reports and 9 proceedings. Out of those papers, four were published as **Physical Review Letters** and two publications were selected as EDITORS' SUGGESTION (in Phys. Rev. Lett. and Phys. Rev. D). In addition, I was a co-editor for two community reports of topical workshops and I co-wrote a review on *Thermalization in QCD* for **Reviews of Modern Physics**.

2020

1. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Predicting parton energy loss in small collision systems*, arXiv:2007.13758
2. A. Huss, A. Kurkela, A. Mazeliauskas, R. Paatelainen, W. van der Schee, U. Wiedemann, *Discovering partonic rescattering in light nucleus collisions*, arXiv:2007.13754
3. J. Berges, M.P. Heller, A. Mazeliauskas, R. Venugopalan, *Thermalization in QCD: theoretical approaches, phenomenological applications, and interdisciplinary connections*, arXiv:2005:12299
4. EDITOR FOR J. Adolfsson et al., *QCD Challenges from pp to A-A Collisions*, community report, arXiv:2003.10997
5. EDITOR FOR M. Bluhm et al., *Dynamics of critical fluctuations: Theory – phenomenology – heavy-ion collisions*, community report, **Nucl. Phys. A** 1003, 122016 (2002), arXiv:2001.08831

2019

6. O. Garcia-Montero, N. Löher, A. Mazeliauskas, J. Berges, and K. Reygers, *Probing the evolution of heavy-ion collisions using direct photon interferometry*, **Phys. Rev. C** 102, 024915 (2020), arXiv:1909:12246

7. D. Devetak, A. Dubla, S. Floerchinger, E. Grossi, A. Mazeliauskas, S. Masciocchi, and I. Selyuzhenkov, *Global fluid fits to identified particle transverse momentum spectra from heavy-ion collisions at the Large Hadron Collider*, **JHEP** 2006 (2020) 44 arXiv:1909:10485
8. P. Hanus, A. Mazeliauskas, and K. Reygers, *Entropy production in pp and Pb-Pb collisions at energies available at the CERN Large Hadron Collider*, **Phys. Rev. C** 100, 064903 (2019), arXiv:1908.02792
9. G. Giacalone, A. Mazeliauskas and S. Schlichting, *Hydrodynamic Attractors, Initial State Energy, and Particle Production in Relativistic Nuclear Collisions*, **Phys. Rev. Lett.** 123, 262301 (2019), arXiv:1908.02866
10. A. Mazeliauskas and V. Vislavicius, *Temperature and fluid velocity on the freeze-out surface from π, K, p spectra in pp, p–Pb and Pb–Pb collisions*, **Phys. Rev. C** 101, 014910 (2020), arXiv:1907.11059
11. W. Florkowski, A. Kumar, R. Ryblewski, and A. Mazeliauskas, *Longitudinal spin polarization in a thermal model*, **Phys. Rev. C 100**, 054907 (2019), arXiv:1904.00002

2018

12. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in weakly coupled QCD*, **Phys. Rev. D** 99, 054018 (2019), EDITORS' SUGGESTION, arXiv:1811.03068
13. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in hadronic collisions*, **Phys. Rev. Lett.** 122, 142301 (2019), arXiv:1811.03040
14. A. Mazeliauskas, J. Berges, *Prescaling and far-from-equilibrium hydrodynamics in the quark-gluon plasma*, **Phys. Rev. Lett.** 122, 122301 (2019), arXiv:1810.10554
15. A. Mazeliauskas, S. Floerchinger, E. Grossi, D. Teaney, *Fast resonance decays in nuclear collisions*, **Eur. Phys. J. C** (2019), arXiv:1809.11049
16. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory*, **Phys. Rev. Lett.** 122, 122302 (2019), EDITORS' SUGGESTION, arXiv:1805.01604
17. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Effective kinetic description of event-by-event pre-equilibrium dynamics in high-energy heavy-ion collisions*, **Phys. Rev. C,99**, 034910 (2019), arXiv:1805.00961

pre-2018

18. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *Bulk viscosity from hydrodynamic fluctuations with relativistic hydro-kinetic theory*, **Phys. Rev. C** 97, 024902 (2018), arXiv:1708.05657
19. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, **Phys. Rev. C**,95, 014909 (2017), arXiv:1606.07742
20. L. Keegan, A. Kurkela, A. Mazeliauskas and D. Teaney, *Initial condition for hydrodynamics from weakly coupled pre-equilibrium evolution*, **JHEP** 1608 (2016) 171, arXiv:1605.04287
21. A. Mazeliauskas and D. Teaney, *Fluctuations of harmonic and radial flow in heavy ion collisions with principal components*, **Phys. Rev. C** 93, 024913 (2016), arXiv:1509.07492
22. A. Mazeliauskas and D. Teaney, *Subleading harmonic flows in hydrodynamic simulations of heavy ion collisions*, **Phys. Rev. C** 91, 044902 (2015), arXiv:1501.03138

Conference proceedings

1. A. Kurkela and A. Mazeliauskas, **SQM 2019**, *Chemical equilibration of QGP in hadronic collisions*, to be published in Springer Proc. Phys., arXiv:1910.06664.

2. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **Confinement XIII**, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory,* PoS Confinement2018 (2019), 152
3. A. Mazeliauskas, **Quark Matter 2018** *Initial conditions for nuclear collisions: theory overview*, Nucl. Phys. A 982 (2019), 134-141, arXiv:1807.05586
4. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **HPT 2017**, *Initial conditions for heavy ion collisions with QCD kinetic theory*, PoS High-pT2017 (2019), 001
5. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **HP2018**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, PoS HardProbes2018 (2018), 111
6. A. Mazeliauskas, Y. Akamatsu and D. Teaney, **CPOD 2017**, *Out-of-equilibrium hydrodynamic fluctuations in the expanding QGP*, PoS CPOD2017 (2018), 038
7. Y. Akamatsu, A. Mazeliauskas and D. Teaney, **Quark Matter 2017**, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, Nucl. Phys. A 967 (2017), 872-875, arXiv:1705.08199 (talk presented by Y. Akamatsu)
8. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **Quark Matter 2017**, *Initial conditions for hydrodynamics from kinetic theory equilibration*, Nucl. Phys. A 967 (2017), 289-292, arXiv:1704.05242
9. A. Mazeliauskas, **HQ2016**, *Initial conditions for hydrodynamics from weakly coupled pre-equilibrium evolution*, J. Phys. Conf. Ser. 832 (2017) no.1, 012016, arXiv:1610.09912

PhD thesis

A. Mazeliauskas, *Fluctuations in ultra-relativistic heavy ion collisions*, arXiv:1702.01202, Max Dresden Prize for outstanding theoretical thesis, degree awarded on 19/05/2017